

Week 05 Face-to-Face Class Outline

Topic: Connected Smart Objects

1. Smart object composition
 - a. Sensors and/or actuators
 - b. Communication device
 - c. Power source
 - d. Processing unit that coordinates all components and processes data
2. Trend of smart objects
 - a. Decreasing size
 - b. Increasing processing power/capability
 - c. Improved communication capabilities
 - d. Standardized communication technologies
3. Sensor network
 - a. A network of smart sensors that can collaborate to accomplish a certain task
 - b. Sensor operation can be either event-driven or period
4. Advantages of networking smart objects
 - a. Extend sensing range
 - b. Improve fault tolerance because the smart objects can backup each other
 - c. Distributed processing of sensing data helps deal with the big sensor data challenge
 - d. Scale better
 - i. Task completion quality and be trade for system lifetime, e.g., by duty cycling the smart objects in the network
 - e. Teamwork to complete larger tasks
5. Communication range
 - a. Short-range (tens of meters): Bluetooth
 - b. Medium-range (tens to hundreds of meters): ZigBee, 802.15.4, low-power Wi-Fi
 - c. Long-range (kilometers): 2G-5G cellular systems
6. Topology
 - a. Star, clustered stars, mesh, peer-to-peer
7. Zigbee: A low-power wireless protocol stack
 - a. 802.15.4 defines PHY (physical) and MAC (media access control) layers of Zigbee

- b. Raw Zigbee doesn't include transport layer. However, by employing an adaptation layer, Zigbee is extended to Zigbee IP protocol stack. (Related topic: "Adaptation" in Week 07 recorded lecture)
- c. At the 2.4GHz unlicensed frequency, Zigbee can achieve 250kbps data rate
- d. At the sub-GHz unlicensed frequency (e.g., 915 MHz and 868 MHz), Zigbee has reduced data rate but increased communication range, compared with 2.4GHz
- e. Zigbee supports Advanced Encryption Standard (AES)-based security. More specifically, it encrypts data to protect confidentiality

8. Low-power WiFi

- a. 802.11ah defines PHY and MAC layers of low-power WiFi
- b. Uses sub-GHz for better signal penetration and longer communication range
- c. Supports nearly 10,000 IoT devices per access point
- d. Longer communication range than standard WiFi
- e. Data rate: 100 kbps (compare this with Zigbee's 250kbps @ 2.4GHz and think about the reason)